

CJ Geier

Dr. Wang

Seminar In Data Analytics

September 18th, 2026

Evaluating the Effect of Play-Action Passing on Defensive Disruption: A Literature Review

Football has drastically changed over time as decades of rule changes and adaptations have evolved the game to something unrecognizable from its earliest years. Specifically, over the past two decades, football has shifted away from run-heavy offenses toward explosive passing schemes. This usually resulted in a high frequency of drop-back passing, usually from the shotgun formation. However, in the last five years offenses have begun to use a run fake known as “play-action” to become even more efficient. As defined by Eager and Seth, play-action passes are plays where the offense fakes a handoff to a ball carrier before the quarterback drops back to attempt a throw (Eager & Seth, 2023, 3). The goal of this is to stop the front seven defenders from pass rushing, and pull linebackers forward against the simulated “run”, opening space behind them to make easier throws in the intermediate part of the field. While it is well documented that play-action increases offensive production (Sharp, 2019, 6), existing research never really explains the “why”. Most of the studies stop at basic stats such as yardage or Expected Points Added (EPA) (Thrush, 2021, 7). However, those metrics cover the impact only at a surface level, and gloss over the critical first second after the snap, where the actual deception happens, explaining the “why”.

Relying only on these basic stats creates a blind spot when trying to evaluate offensive play calling and the true impact play-action has. When just looking at the final yardage or EPA

of a play, it is impossible to tell if play-action worked because of actual hesitation or confusion by the defense, or because offenses usually call it in advantageous short-distance situations. To address this, my project will take advantage of tracking data available from the NFL Big Data Bowl (National Football League, 2024, 5) to truly see what happens at the snap, looking at metrics like defensive line get-off speed, pressure time, and defender displacement. I will also match situations, allowing me to control for factors like down, distance, and field position to better find the true impact of the play-action. On top of that, I also want to see if running a lot of play-action has a positive effect on normal running plays. Using the tracking metrics, I will see if defenses start playing more passively, increasing offensive efficiency purely through the threat of an effective play-action attack. By matching the situations and evaluating the tracking data at the snap, I will hopefully be able to start to see how much of play-action's success is truly caused by defensive disruption.

Calling plays for an offense comes down to a makeshift chess match between the offensive coordinator, who is usually the person who decides which play is called, and the defense. For the early downs (1st and 2nd down), football has even been looked at through game theory by comparing it to the matching pennies game (McGarrity & Linnen, 2010, 4). In the setup example they used, the offense picks between a run and a pass, while the defense would commit its personnel to stop one or the other. If the defense guesses run, linebackers will trigger downhill, and the defense will crowd the line of scrimmage. If they guess pass, the pass rushers will get a faster get-off time and linebackers drop into their zones faster. This is a great example of how important it is for offenses to stay unpredictable so defenses cannot correctly predict what they are going to call. Even though this is clear, stats show that NFL coaches still often leave value on the table on early downs. By looking at first-down data using Generalized Additive

Models, which use flexible curves to handle non-linear situations like field position and down. It was proven that throwing the ball is actually the better analytical choice over 80% of the time (Wilk, 2026, 8). But yet offenses still run the ball more often than they throw it on first down. Though this may be because of other factors such as risk aversion and fear of turnovers, teams seem to be leaving points on the table. This gap between what coaches actually do and what the analytics say they should do is where play-action proves to be extremely effective. On these early downs, defenses naturally expect the offense to run because that is what coaches have traditionally called for a majority of the history of the NFL. When an offense then fakes the handoff and uses play-action, they not only get to take advantage of the defense expecting a run, but also get to pass the ball, which is proven to be more efficient. This also makes these passes even more effective than normal ones, as the defense's reaction to the pass is much slower than it normally is to a regular passing play, opening up larger throwing windows in the intermediate part of the field.

However, the problem with both the work of McGarrity and Linnen, and Wilk's is how they measured these plays. Both of the studies depended entirely on box score stats and final EPA values, completely disregarding the tracking data I will be using. When you only look at the final play result, you miss a larger part of the picture and the effect that play-action is having on the defense. Since a final box score number cannot tell you if a linebacker hesitated for a second or took a step in the wrong direction due to the play fake, which is the exact physical reaction that makes play-action work in the first place.

To actually measure what happens in that first second after the snap, new research has moved away from traditional box scores and turned to player tracking data that is becoming more available than ever before. The standard data for this comes from the NFL Big Data Bowl,

which provides raw coordinate tracking at ten frames per second using RFID chips in players' shoulder pads. Though it is important to note that working with this kind of data in R can be very difficult and usually requires heavy data manipulation with packages like Dplyr to filter down the noise that comes with the data. Some other researchers have handled this by focusing on the front seven defenders through measuring defensive line get-off speed and linebacker displacement (Eager et al., 2022, 2). They used Euclidean distance formulas to track how far linebackers stepped toward the line of scrimmage during play-action, creating a metric for linebackers called "bite distance" (Eager et al., 2022, 2). They also tracked pass rusher get-off time by measuring the time it took for defensive linemen to cross the line of scrimmage (Eager et al., 2022, 2). This approach is similar to how I will be measuring the effects of defensive disruption, however, I will have access to a much more recent and expansive database. I will also be using more metrics that add more context to offensive efficiency, such as EPA and defensive pressure rate.

Another area analysts have looked into is how quickly defenders are forced to change direction post-snap when reacting to play-action. This was determined by comparing the direction a defender is facing to the actual direction they are running (Attrah, 2026, 1). When a linebacker bites on play-action, they have to completely stop their forward momentum and flip their hips to retrace back to their coverage drop responsibility. When there is a big difference between where a defender is facing and where they need to go, their acceleration is much slower (Attrah, 2026, 1). This can be used to show why even a small step in the wrong direction can completely take a defender out of a passing lane and result in big plays for the offense. While that research explains the physical cost of turning around, my project aims to take this concept

further by connecting it directly to pressure rates and the downtick in defensive efficiency that results from this change in direction.

Looking at all the research that is being done on play-action, it is clear that while there has been plenty of work done, a lot of it misses the mark on the things that coaches, players, and everyday fans can actually understand and use. Most existing studies do a great job of proving that play-action is effective, but fall short of explaining the actual "why," which is so key to the teaching part of football that helps evolve the game at all levels. That gap is why combining the traditional metrics with the advanced tracking data and analytics will paint a much cleaner picture. Reviewing this literature has also helped me figure out how to practically attack such a dense and massive dataset. Working with tracking coordinates is considered very difficult and can become very overwhelming, especially with a clear timeline. But seeing how past studies used R packages like SportyR for plotting and SP to calculate player distance formulas gave me a good idea of how I can attack this data and keep it realistic for this semester. One of my goals for this research is also to help influence offensive game plans at the high school and college level. It usually takes years for NFL trends to trickle down to lower levels of football, and I hope that with more practical research like this readily available, it could help expedite that process. By bridging the gap between player coordinates and the base stats, while controlling for situations, this project will provide a clean answer to how play-action disrupts defenses and increases offensive efficiency.

References

1. Attrah, J. (2026). *Kinematic tracking and directional change in NFL defender movement*. Journal of Sports Analytics.
2. Eager, E., Chahrouri, G., & PFF Research. (2022). *Quantifying front-seven reaction to play-action using computer vision and tracking data*. Pro Football Focus Research.
3. Eager, E., & Seth, T. (2023). *The mechanics and efficiency of play-action passing in modern NFL offenses*. SumerSports.
4. McGarrity, J. P., & Linnen, B. (2010). Pass or run: An empirical test of the matching pennies game in the National Football League. *Southern Economic Journal*, 77(1), 102–110.
5. National Football League. (2024). *NFL Big Data Bowl 2024: Player tracking data and analytics*. NFL Football Operations.
6. Sharp, W. (2019). *Play-action passing efficiency and defensive run commitment in professional football*. Sharp Football Analysis.
7. Thrush, K. (2021). *Evaluating the limitations of expected points added in sub-second play-calling deception*. Journal of Quantitative Analysis in Sports.
8. Wilk, A. (2026). *Modeling early-down NFL play-calling efficiency using Generalized Additive Models*. MIT Sloan Sports Analytics Conference.

Note: References follow APA 7th edition formatting guidelines.